Executive Summary

Social Media Analytics Independent Study: Corporate Communication on LGBTQ Issues

Background

This independent study investigates how corporations communicate about LGBTQ-related issues on social media, with a particular focus on Twitter (X). Building upon previous research examining corporate engagement with LGBTQ topics, the project aims to develop a more comprehensive understanding of corporate communication strategies by combining traditional exploratory data analysis with modern Natural Language Processing (NLP) techniques.

While previous work primarily identified LGBTQ-related tweets and measured overall sentiment, this study extends the analysis by examining the nature of corporate messaging, the sentiment expressed within those messages, and the extent to which corporate communication overlaps with broader political or public policy discussions.

The project utilizes a large dataset of corporate tweets collected from publicly traded companies over multiple years

Research Objectives

The project seeks to answer several related research questions:

- How has corporate LGBTQ communication evolved over time?
- To what extent is corporate LGBTQ communication concentrated around Pride Month?
- What types of LGBTQ communication are companies producing?
- How does sentiment vary across LGBTQ-related corporate messaging?
- How political is corporate social media communication, both within LGBTQ discussions and across all corporate tweets?

To address these questions, the study combines descriptive analytics with transformer-based zero-shot classification and VADER sentiment analysis.

Methodology

The project was divided into four complementary analytical components.

1. Exploratory Data Analysis

The first stage focused on understanding the overall structure of the dataset and identifying temporal trends in corporate LGBTQ communication.

This analysis included:

- dataset validation and cleaning
- yearly tweet volumes
- company-level tweet distributions
- engagement metrics
- Pride Month trend analysis
- comparison of June versus non-June communication
- yearly percentage of LGBTQ tweets relative to all corporate tweets
- comparison of communication before, during, and after Pride Month
- additional analyses excluding Netflix, whose entertainment-related LGBTQ content differs substantially from typical corporate messaging.

These analyses establish a descriptive foundation for the remainder of the project and provide insight into how corporate engagement with LGBTQ issues has changed over time.

2. Sentiment Analysis

The second component evaluated the emotional tone of LGBTQ-related corporate communication using the VADER sentiment analysis model.

Rather than simply measuring whether tweets were positive or negative, this analysis explored how sentiment scores relate to the language used within corporate messaging.

Preliminary findings indicated that negative sentiment occurred more frequently than initially expected. Manual inspection suggested that many of these tweets were not expressing opposition toward LGBTQ communities, but instead discussed discrimination, violence, legislation, or hate speech using language that VADER naturally classified as negative.

This highlights an important methodological consideration when interpreting lexicon-based sentiment analysis on social issues, where negative wording may actually accompany supportive corporate messaging.

3. Zero-Shot Topic Classification

To better understand the content of LGBTQ-related communication, the study employs transformer-based zero-shot classification.

Unlike traditional topic modeling approaches, zero-shot classification allows tweets to be categorized without manually labeled training data.

The classification framework assigns independent confidence scores for four communication dimensions:

- Support
- Company Action
- Employee Profile
- Political Statement

Unlike mutually exclusive classification, each tweet receives a confidence score for all four categories, allowing a single tweet to simultaneously express support while describing company initiatives or employee experiences.

This probabilistic approach provides a richer representation of corporate communication than assigning a single category to each tweet.

The resulting classifications allow the project to examine:

- company communication strategies
- temporal changes in communication styles
- engagement differences across communication types
- examples of high-confidence tweets within each category

4. Political Content Classification

A separate zero-shot classification pipeline was developed to measure the extent to which corporate communication contains political or public policy discussion.

Unlike the LGBTQ categorization notebook, this analysis operates on the complete corporate tweet dataset rather than only LGBTQ-related tweets.

The notebook was intentionally designed to support two execution modes.

A development mode randomly samples a subset of tweets for testing and exploratory analysis, while a production mode allows the same workflow to be executed on the entire dataset by simply disabling the sampling step.

This modular design enables efficient experimentation while remaining scalable for large-scale analysis.

The resulting political probability scores can later be compared against engagement metrics, company behavior, and temporal trends.

Preliminary Insights

The exploratory analysis indicates that corporate LGBTQ communication is strongly concentrated around Pride Month, although meaningful variation exists across companies and years.

The sentiment analysis demonstrates that negative sentiment scores should be interpreted carefully, as they often reflect discussions of discrimination or social issues rather than negative corporate attitudes toward LGBTQ communities.

The zero-shot classification framework provides a more nuanced understanding of corporate messaging by distinguishing between expressions of support, descriptions of organizational initiatives, employee-focused communication, and political discussion.

Together, these analyses move beyond measuring the volume of LGBTQ communication and begin examining the content and purpose of corporate messaging.

Future Work

The analytical framework developed in this study also creates several opportunities for further research. Future work could examine how different forms of corporate LGBTQ communication vary across industries, companies, and time periods, and whether particular communication strategies are associated with higher levels of audience engagement. The relationship between sentiment and communication type could also be explored in greater depth, particularly to distinguish negative language describing social issues from negative attitudes toward LGBTQ communities.